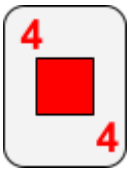
American Computer Science League

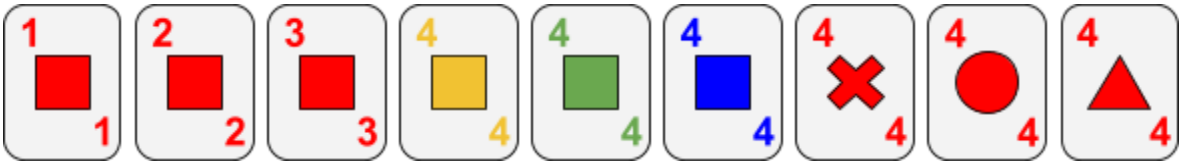
2024-2025 ● Contest 2: Uno Duo ● Junior Division



PROBLEM: The card game *Duo* uses cards that have 3 characteristics: a color, number, and symbol. The game starts with two face-up cards and 7 cards in the player's hand. On a single turn, the player can choose one of the two face-up cards and *match* as many cards from their hand as possible on that pile. A card can be placed on top of another card if exactly 2 of the characteristics match.



The ACSL cards use four colors (R=red, Y=yellow, G=green, B=blue), four numbers (1, 2, 3, 4), and four symbols (X=cross, O=circle, S=square, T=triangle). A card is represented by a 3-character string showing the color, number, and symbol, in that order. For example, the card R4S, shown on the left, is red with the number 4 and contains a square. There are 9 cards that could possibly be placed on top of this card: a red square with the number 1, 2, or 3; a yellow, green, or blue square with the number 4; a red cross, circle, or triangle with the number 4.



In this version of the game called *Uno Duo*, your program will play one turn for a single player on a single pile. The turn begins with one face-up card and 7 cards in the hand (think of them as being held from left to right). The following steps make up a turn:

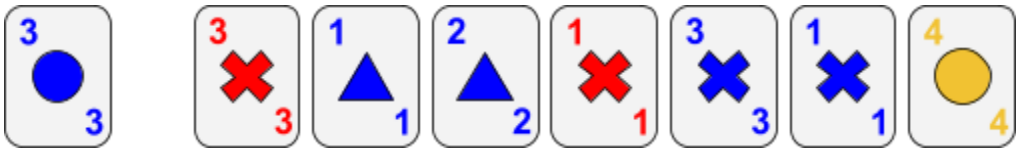
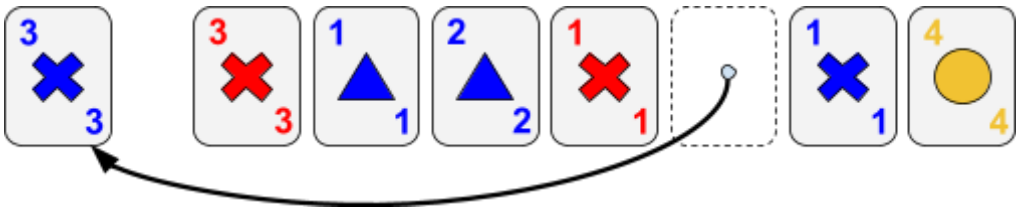
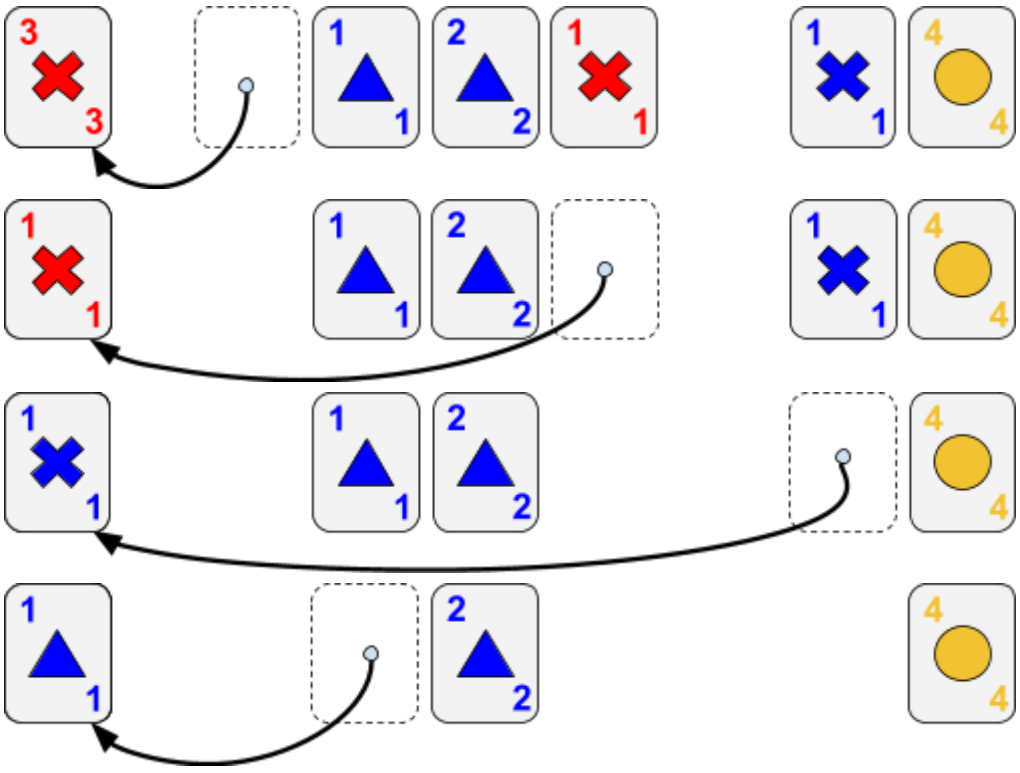
1. Find the leftmost card in your hand that matches the face-up card. Place the matched card on the top, thus creating a face-up pile of cards.
2. Try to match as many cards in your hand to the top card on the pile. Each time, look at your cards starting at the leftmost. If the card is a match, place it on top of the pile of cards.
3. Stop when you cannot play any more cards in your hand.

The output is a string containing all of the cards on the pile, from bottom to top, each separated by a single space.

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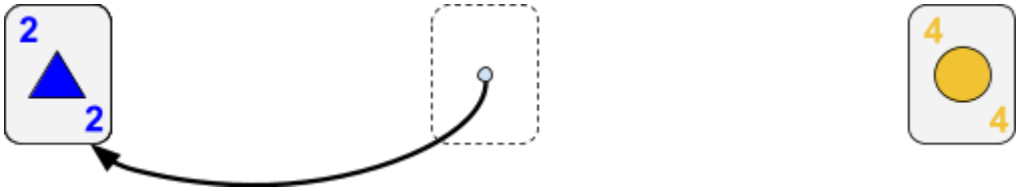
2024-2025 ● Contest 2: Uno Duo ● Junior Division

EXAMPLE:

Input	`B3O` `R3X B1T B2T R1X B3X B1X Y4O`
Output	`B3O B3X R3X R1X B1X B1T B2T`
Explanation	Initially, the face-up card and the 7 cards in the hand are as follows:  The first card in the hand that can be played is B3X (matching the color and the number, but not the shape) on the face-up card, thus making a pile.  The remaining moves are as follows: 

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At this point, the only card remaining in the hand is Y4O. It cannot be placed on the pile so the turn ends. The output is the string containing the cards in the pile, from bottom to top: `B3O B3X R3X R1X B1X B1T B2T`.

INPUT: There will be two strings. The first string is the face-up card. The second string is the 7 cards in the player's hand. All cards are separated by a single space. There will be no duplicate cards in each data set.

OUTPUT: Output a string containing all of the cards in the pile from bottom to top, each separated by a single space.

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2024-2025 ● Contest 2: Uno Duo ● Junior Division

SAMPLE INPUT:

```
1. B3O
   R3X B1T B2T R1X B3X B1X Y4O
2. R4S
   R4X B4X B3X B3T G3T G1T R1O
3. B1T
   G1O B1O R1T B4T G4O R4O R1X
4. G3S
   R3S G1O B4X G2S Y4S G3T G3X
5. Y4O
   Y1S G4S B4S G2S R1T G2X Y2T
6. G2T
   Y1S G4S B4S G2S R1T G2X Y2T
```

SAMPLE OUTPUT:

```
1. B3O B3X R3X R1X B1X B1T B2T
2. R4S R4X B4X B3X B3T G3T G1T
3. B1T B1O G1O G4O R4O
4. G3S R3S
5. Y4O
6. G2T G2S G4S B4S
```

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TEST DATA

TEST INPUT:

1. `Y2S`
   `B2S Y1X G2S R4S G1O Y4T G2T`
2. `R1S`
   `G4S B1T G1T B4S Y2X Y1S Y1T`
3. `R2X`
   `Y2T Y1T G2T B4O G1X R4X B3S`
4. `G1O`
   `R1O B3S G1X R3O R4O R4T B3X`
5. `R1S`
   `G1S B1T G1T Y4S Y2S Y1S Y1T`
6. `Y4O`
   `G1X Y1O Y1X R2X B4S B4O R1X`

TEST OUTPUT:

1. `Y2S B2S G2S G2T`
2. `R1S Y1S Y1T B1T G1T`
3. `R2X R4X`
4. `G1O R1O R3O R4O R4T`
5. `R1S G1S G1T B1T Y1T Y1S Y4S Y2S`
6. `Y4O Y1O Y1X G1X R1X R2X`

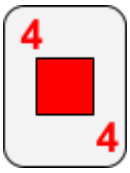
American Computer Science League

2024-2025 ● Contest 2: Uno Duo ● Junior Division

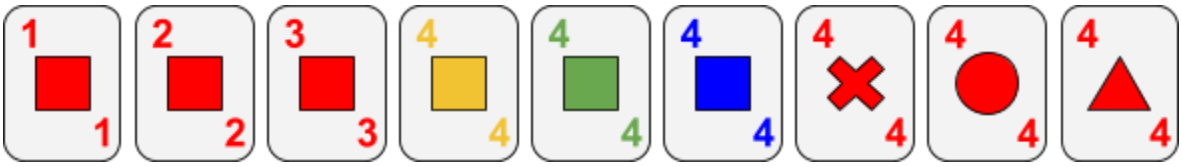
PROBLEM STATEMENT:



The card game *Duo* uses cards that have 3 characteristics: a color, number, and symbol. The game starts with two face-up cards and 7 cards in the player's hand. On a single turn, the player can choose one of the two face-up cards and *match* as many cards from their hand as possible on that pile. A card can be placed on top of another card if exactly 2 of the characteristics match.



The ACSL cards use four colors (R=red, Y=yellow, G=green, B=blue), four numbers (1, 2, 3, 4), and four symbols (X=cross, O=circle, S=square, T=triangle). A card is represented by a 3-character string showing the color, number, and symbol, in that order. For example, the card R4S, shown on the left, is red with the number 4 and contains a square. There are 9 cards that could possibly be placed on top of this card: a red square with the number 1, 2, or 3; a yellow, green, or blue square with the number 4; a red cross, circle, or triangle with the number 4.



In this version of the game called *Uno Duo*, your program will play one turn for a single player on a single pile. The turn begins with one face-up card and 7 cards in the hand (think of them as being held from left to right). The following steps make up a turn:

1. Find the leftmost card in your hand that matches the face-up card. Place the matched card on the top, thus creating a face-up pile of cards.
2. Try to match as many cards in your hand to the top card on the pile. Each time, look at your cards starting at the leftmost. If the card is a match, place it on top of the pile of cards.
3. Stop when you cannot play any more cards in your hand.

The output is a string containing all of the cards on the pile, from bottom to top, each separated by a single space.

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2024-2025 ● Contest 2: Uno Duo ● Junior Division

EXAMPLE:

Input

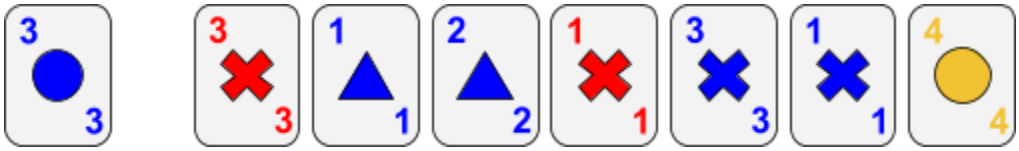
```
B3O
R3X B1T B2T R1X B3X B1X Y4O
```

Output

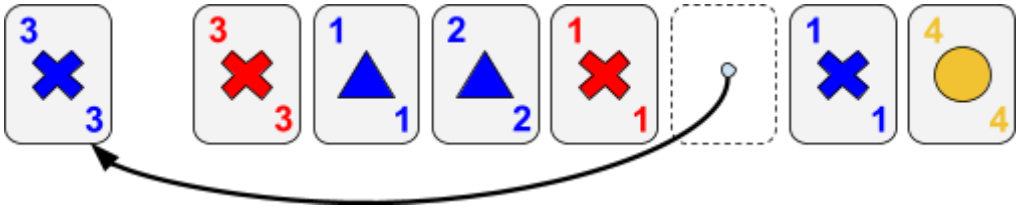
```
B3O B3X R3X R1X B1X B1T B2T
```

Explanation

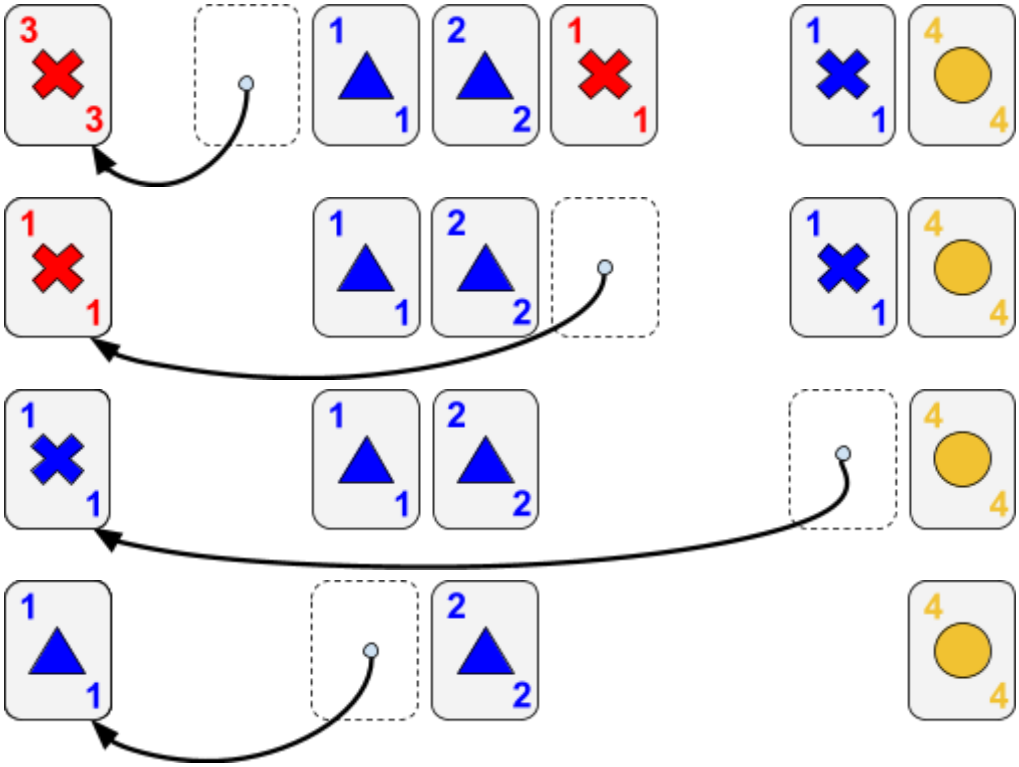
Initially, the face-up card and the 7 cards in the hand are as follows:



The first card in the hand that can be played is B3X (matching the color and the number, but not the shape) on the face-up card, thus making a pile.

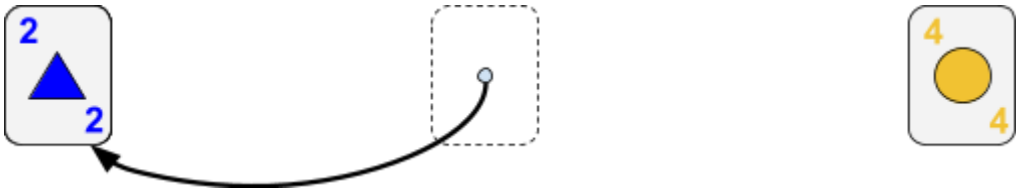


The remaining moves are as follows:



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At this point, the only card remaining in the hand is Y4O. It cannot be placed on the pile so the turn ends. The output is the string containing the cards in the pile, from bottom to top: `B3O B3X R3X R1X B1X B1T B2T.`

TASK:

Complete the function **printPile**

- The function has 2 parameters, both strings: the first, *pile*, representing the face-up card and the second, *hand*, representing the 7 cards in a player's hand.
- The function returns a string containing all of the cards in the pile from bottom to top, each card separated by a single space.

You may create additional functions that are called from **printPile** if needed in solving the problem.

CONSTRAINTS:

There will be one face-up card and 7 cards in a player's hand. There will be no duplicate cards in each data set.

DATA PROVIDED:

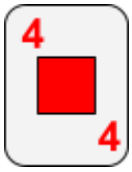
There are 6 sets of Sample Data for debugging and 6 sets of Test Data for scoring. The test cases will vary in difficulty. You should create sample data of your own to fully test your program.

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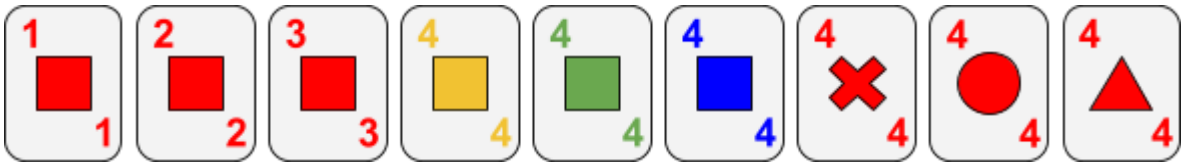
2024-2025 ● Contest 2: Uno Duo ● Junior Division

问题：*Duo* 卡牌游戏中，每张卡牌有 3 个特征：颜色、数字和符号。游戏开始时，有两张卡牌正面朝上，且玩家手中持有 7 张卡牌。在一轮游戏中，玩家可以从两张正面朝上的卡牌中选择一张，并尽可能多地将手中的卡牌与该牌堆上的卡牌进行*匹配*。如果两张卡牌恰好有两个特征相匹配，就可以将一张卡牌放在另一张卡牌的上面。





ACSL 版卡牌使用四种颜色（R = 红色，Y = 黄色，G = 绿色，B = 蓝色）、四个数字（1、2、3、4）和四种符号（X = 叉号，O = 圆形，S = 正方形，T = 三角形）。每张卡牌用一个包含 3 个字符的字符串表示，并按照颜色、数字、符号的顺序依次排列。例如，左侧所示的卡牌 R4S 是一张颜色为红色，数字为 4，符号为正方形的卡牌。有 9 张卡牌可能会被放在这张卡牌的上面：颜色为红色，数字为 1、2 或 3，符号为正方形的卡牌；颜色为黄色、绿色或蓝色，数字为 4，符号为正方形的卡牌；颜色为红色，数字为 4，符号为叉号、圆形或三角形的卡牌。



在这个名为 *Uno Duo* 的游戏版本中，你的程序将模拟单个玩家在一轮游戏中对单个牌堆的操作。这一轮游戏开始时，有一张卡牌正面朝上，且玩家手中持有 7 张卡牌（可将卡牌顺序视为从左至右排列）。一轮游戏的操作按以下步骤进行：

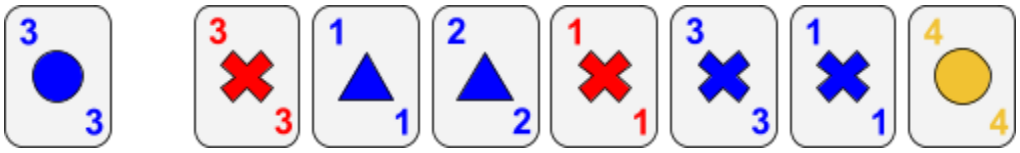
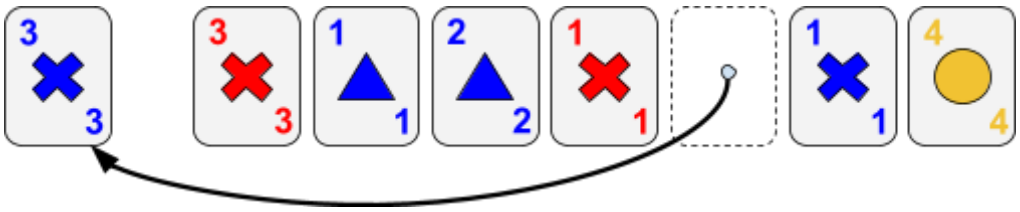
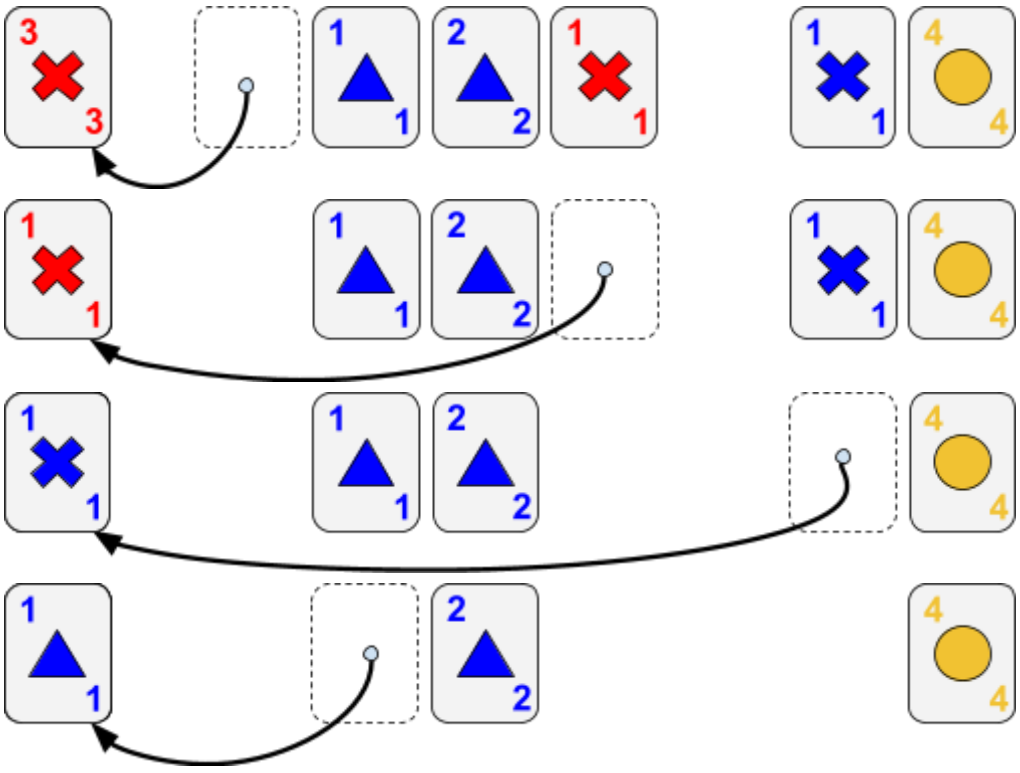
1. 从手牌中找到最左侧一张与正面朝上的卡牌相匹配的牌。将匹配的卡牌放在最上面，从而形成一个正面朝上的牌堆。
2. 尝试将手中尽可能多的卡牌与牌堆最上面的卡牌匹配。每次都从最左侧的卡牌开始查看。如果卡牌匹配，就将其放在牌堆最上面。
3. 当手中的卡牌无法再继续匹配时，游戏结束。

输出为一个字符串，包含牌堆中从底部到顶部的所有卡牌，每张卡牌之间用一个空格隔开。

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2024-2025 ● Contest 2: Uno Duo ● Junior Division

样例：

输入	B3O R3X B1T B2T R1X B3X B1X Y4O
输出	B3O B3X R3X R1X B1X B1T B2T
解释说明	游戏开始时，正面朝上的卡牌和玩家手中持有的 7 张卡牌如下所示：  手中可以匹配的第一张卡牌是 B3X（颜色和数字匹配，但符号不匹配），将其放在正面朝上的卡牌上面，从而形成一个牌堆。  接下来的卡牌匹配情况如下所示： 

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样例输入：

```
1. B3O
   R3X B1T B2T R1X B3X B1X Y4O
2. R4S
   R4X B4X B3X B3T G3T G1T R1O
3. B1T
   G1O B1O R1T B4T G4O R4O R1X
4. G3S
   R3S G1O B4X G2S Y4S G3T G3X
5. Y4O
   Y1S G4S B4S G2S R1T G2X Y2T
6. G2T
   Y1S G4S B4S G2S R1T G2X Y2T
```

样例输出：

```
1. B3O B3X R3X R1X B1X B1T B2T
2. R4S R4X B4X B3X B3T G3T G1T
3. B1T B1O G1O G4O R4O
4. G3S R3S
5. Y4O
6. G2T G2S G4S B4S
```

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测试数据

测试输入：

1. Y2S
   B2S Y1X G2S R4S G1O Y4T G2T
2. R1S
   G4S B1T G1T B4S Y2X Y1S Y1T
3. R2X
   Y2T Y1T G2T B4O G1X R4X B3S
4. G1O
   R1O B3S G1X R3O R4O R4T B3X
5. R1S
   G1S B1T G1T Y4S Y2S Y1S Y1T
6. Y4O
   G1X Y1O Y1X R2X B4S B4O R1X

测试输出：

1. Y2S B2S G2S G2T
2. R1S Y1S Y1T B1T G1T
3. R2X R4X
4. G1O R1O R3O R4O R4T
5. R1S G1S G1T B1T Y1T Y1S Y4S Y2S
6. Y4O Y1O Y1X G1X R1X R2X

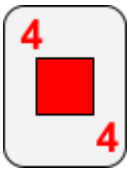
American Computer Science League

2024-2025 ● Contest 2: Uno Duo ● Junior Division

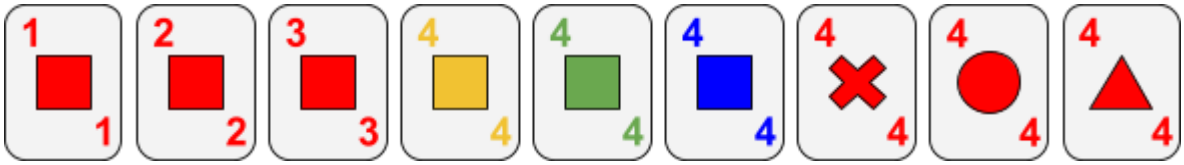
问题陈述：



Duo 卡牌游戏中，每张卡牌有 3 个特征：颜色、数字和符号。游戏开始时，有两张卡牌正面朝上，且玩家手中持有 7 张卡牌。在一轮游戏中，玩家可以从两张正面朝上的卡牌中选择一张，并尽可能多地将手中的卡牌与该牌堆上的卡牌进行*匹配*。如果两张卡牌恰好有两个特征相匹配，就可以将一张卡牌放在另一张卡牌的上面。



ACSL 版卡牌使用四种颜色（R = 红色，Y = 黄色，G = 绿色，B = 蓝色）、四个数字（1、2、3、4）和四种符号（X = 叉号，O = 圆形，S = 正方形，T = 三角形）。每张卡牌用一个包含 3 个字符的字符串表示，并按照颜色、数字、符号的顺序依次排列。例如，左侧所示的卡牌 R4S 是一张颜色为红色，数字为 4，符号为正方形的卡牌。有 9 张卡牌可能可以放在这张卡牌的上面：颜色为红色，数字为 1、2 或 3，符号为正方形的卡牌；颜色为黄色、绿色或蓝色，数字为 4 ，符号为正方形的卡牌；颜色为红色，数字为 4 ，符号为叉号、圆形或三角形的卡牌。



在这个名为 *Uno Duo* 的游戏版本中，你的程序将模拟单个玩家在一轮游戏中对单个牌堆的操作。这一轮游戏开始时，有一张卡牌正面朝上，且玩家手中持有 7 张卡牌（可将卡牌顺序视为从左至右排列）。一轮游戏的操作按以下步骤进行：

1. 从手牌中找到最左侧一张与正面朝上的卡牌相匹配的牌。将匹配的卡牌放在最上面，从而形成一个正面朝上的牌堆。
2. 尝试将手中尽可能多的卡牌与牌堆最上面的卡牌匹配。每次都从最左侧的卡牌开始查看。如果卡牌匹配，就将其放在牌堆最上面。
3. 当手中的卡牌无法再继续匹配时，游戏结束。

输出为一个字符串，包含牌堆中从底部到顶部的所有卡牌，每张卡牌之间用一个空格隔开。

样例：

输入

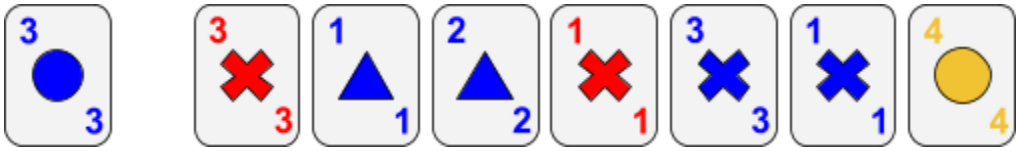
```
B3O
R3X B1T B2T R1X B3X B1X Y4O
```

输出

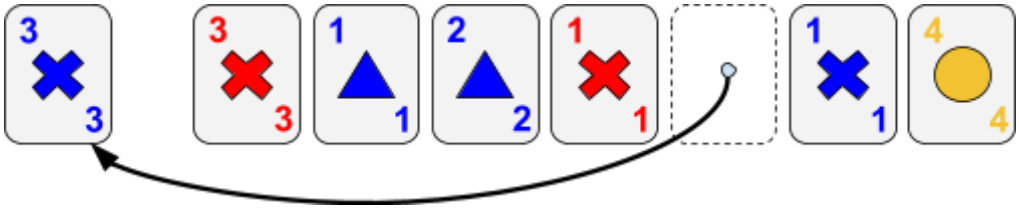
```
B3O B3X R3X R1X B1X B1T B2T
```

解释说明

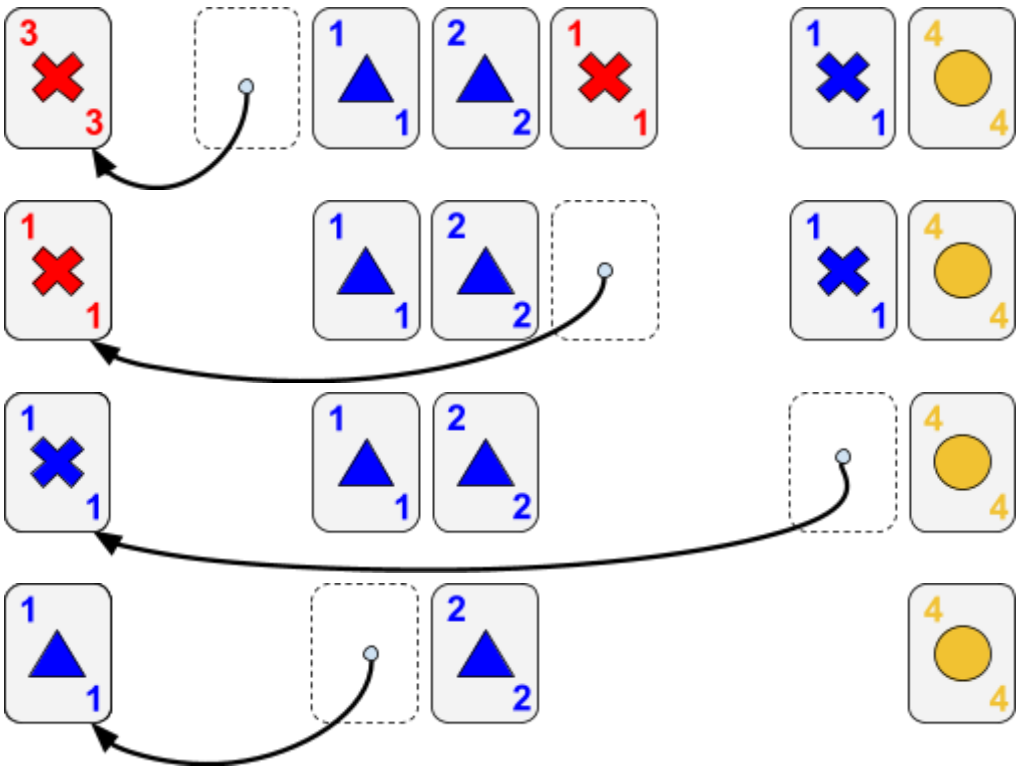
游戏开始时，正面朝上的卡牌和玩家手中持有的 7 张卡牌如下所示：

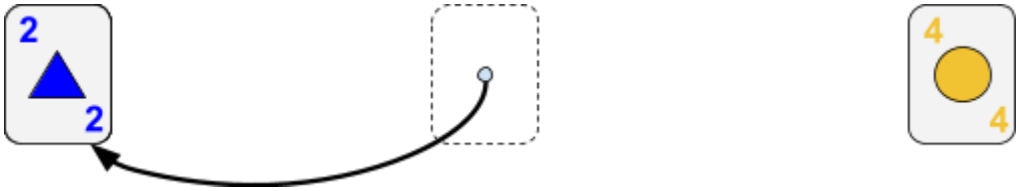


手中可以匹配的第一张卡牌是 B3X（颜色和数字匹配，但符号不匹配），将其放在正面朝上的卡牌上面，从而形成一个牌堆。



接下来的卡牌匹配情况如下所示：





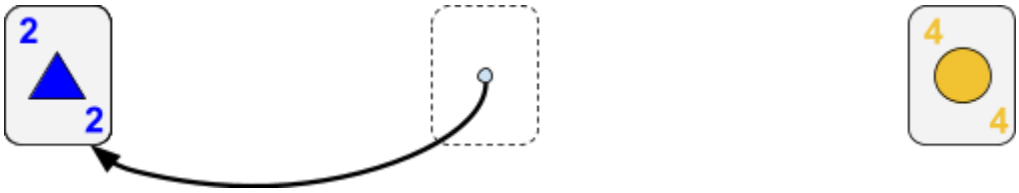
此时，玩家手中唯一剩下的卡牌是 Y4O。该卡牌无法放置在牌堆上，因此本轮游戏结束。输出为一个字符串，包含牌堆中从底部到顶部的卡牌：`B3O B3X R3X R1X B1X B1T B2T.`

输入：将有两个字符串。第一个字符串是正面朝上的卡牌。第二个字符串是玩家手中的 7 张卡牌。所有卡牌之间用一个空格隔开。每个数据集中不会有重复的卡牌。

输出：输出一个字符串，包含牌堆中从底部到顶部的所有卡牌，每张卡牌之间用一个空格隔开。

American Computer Science League

2024-2025 ● Contest 2: Uno Duo ● Junior Division



此时，玩家手中唯一剩下的卡牌是 Y4O。该卡牌无法放置在牌堆上，因此本轮游戏结束。输出为一个字符串，包含牌堆中从底部到顶部的卡牌：`B3O B3X R3X R1X B1X B1T B2T`。

任务：

完成函数 **printPile**

- 该函数有 2 个参数，均为字符串：第一个参数 *pile* 表示正面朝上的卡牌，第二个参数 *hand* 表示玩家手中的 7 张卡牌。
- 函数返回一个字符串，包含牌堆中从底部到顶部的所有卡牌，每张卡牌之间用一个空格隔开。

如有需要，你可以构建额外的函数供 **printPile** 调用来解答此问题。

约束条件：

有一张正面朝上的卡牌和玩家手中的 7 张卡牌。每个数据集中不会有重复的卡牌。

提供的数据：

有 6 组用于调试的样例数据和 6 组用于评分的测试数据。测试用例的难度各不相同。你应该创建自己的样例数据来充分测试你的程序。